

Data Cleaning Techniques

MOST IMPORTANT DATA CLEANING STEPS

- **Removing unnecessary columns:** You can remove any columns that are not needed for your analysis.
 1. Open the Power BI Desktop application.
 2. Load the data set that needs to be cleaned.
 3. Go to the "Transform" button in the home tab. This will take you to Power Query Editor
 4. In the Data Grid, right-click on the column and select the "Remove Columns" option from the context menu.

Make sure to carefully evaluate the data before removing any column, as removing a column may result in the loss of important data.

- **Renaming columns:** You can rename columns to make them more meaningful and easier to understand. In the Power Query Editor, rename columns by double clicking on the column header and typing in the new name
- **Removing duplicate rows:** You can remove duplicate rows to ensure that your data is accurate and free of duplicates.
 1. Go to the "Transform" button in the home tab.
 2. On the top left of the Data Grid, you will see a Grid Icon. Click on it and then select Remove Duplicate from the list. By default, all columns are selected
 3. Click on the "OK" button to remove the duplicates.
- **Merging columns:** You can merge multiple columns into a single column if they contain related information.
 1. Select the columns that you want to merge
 2. In the Transform tab, click on the "Merge Columns" button.
 3. In the Merge Columns dialog box, specify the delimiter (if any).
 4. Click on the "OK" button to merge the columns.
 5. In the Data Grid, you will see the merged column with the specified name.
- **Splitting column values:** You can split a single column into multiple columns if it contains multiple values separated by a delimiter, such as a comma or a space.
 1. Go to Power Query Editor & Select the column which you would like to split
 2. In the Transform tab of the Power Query Editor, Click on "Split Column"
 3. Choose whether you want to split by Delimiter, Character or any other method
 4. Let's choose Delimiter option for now

5. Choose the desired delimiter and instruct whether you want to split at each other occurrence or left/right most delimiter. Press OK
-
- **Replacing null values:** You can replace null or missing values with a default value to ensure that your data is complete and accurate.
 1. Select the column with null values.
 2. Go to the Transform tab and click on the Replace Values function.
 3. In the Replace Values window, enter the value to replace null values within the Replace With field.

 - **Changing data types:** You can change the data type of a column to ensure that it is correctly formatted for your analysis.
 1. In the Data Grid of Power Query Editor, click on the icon on the left side of the column header
 2. This will option a range of options to choose i.e. Decimal Number, Text, Date Time etc..
 3. Choose the desired option and that will change its data type

 - **Sorting data:** You can sort your data by one or more columns to organize it in a meaningful way.
 1. In the Data view of Power Query Editor, click on the drop down next to the column header
 2. Now you can select whether you want to sort the data Ascending or Descending.

 - **Filtering data:** You can filter your data to include only the rows that meet certain criteria, such as a specific date range or a specific value in a particular column.
 1. In the Data view of Power Query Editor, click on the drop down next to the column header
 2. Either enter the value which you want to filter in the Search box or else apply a specific Text or Number filter from the options

 - **Transforming data:** You can use various data transformation techniques, such as pivoting and unpivoting, to reshape your data and make it easier to work with. But this requires a detailed content of its own, so we will park it for now.

Now after all of your cleaning steps are completed, remember to click on Close & Load to apply those changes to your data.